CAMBRIDGE
International Education

Cambridge O Level

PHYSICS **5054/22**

Paper 2 Theory **October/November 2024**

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	‘List rule’ guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Question	Answer	Marks
1(a)(i)	resultant force = increase in momentum / unit time **or** resultant force = rate of increase in momentum	**B1**
1(a)(ii)	(F =) 180 × 30 / 2.0 **or** 180 × 15	**C1**
	2700 (N)	**A1**
1(b)(i)	pump **/** jet ski / engine exerts force on water	**B1**
	water exerts an opposite force on the pump / jet ski / engine	**B1**
1(b)(ii)	(a =) F / m **or** 2700 / (70 + 280) **or** 2700 / 350	**C1**
	7.7 (m / s^2)	**A1**
1(c)	(K.E. =) ½ mv^2	**C1**
	½ × (70 + 280) × 20^2 **or** ½ × 350 × 20^2	**C1**
	70 000 (J)	**A1**

Question	Answer	Marks
2(a)	clear evidence that pressure ∝ 1 / area **or** mg seen	**C1**
	pressure = mg / πr^2	**A1**
2(b)	line with intercept > 0	**B1**
	between t = 0 and t = T, straight line **and** with gradient > 0	**B1**
	horizontal line at end of initial line	**B1**

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Question	Answer	Marks
3(a)	(air next to the pipe contracts / its volume decreases and its) density increases	**B1**
	cold air / air next to the pipe sinks	**B1**
	warm(er) air (at the bottom of the freezing compartment) rises **or** convection (current) set up	**B1**
3(b)(i)	particles vibrate **and** collide with other particles	**B1**
	collisions transfer (thermal) energy / vibration **or** (thermal) energy / vibration passed to neighbouring particles / neighbours / nearby particles	**B1**
3(b)(ii)	air is trapped	**B1**
	(trapping the air / the bubbles / they) reduces / prevents convection	**B1**
	air is a bad conductor	**B1**

Question	Answer	Marks
4(a)(i)	(F =) pA **or** $1.0 \times 10^5 \times 3.5 \times 10^{-3}$	**C1**
	350 (N)	**A1**
4(a)(ii)	force to left **or** force due to atmospheric pressure / pressure of air surrounding the syringe	**B1**
	no resultant force (on piston) **or** (two) forces balance **or** forces in equilibrium	**B1**
4(b)(i)	(particles / they) hit the walls / sides / piston / syringe	**B1**
	(particles / they) less densely packed / more spread out / further apart	**B1**
	collisions (of particles) less frequent	**B1**

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Question	Answer	Marks
4(b)(ii)	(p_2 =) p_1V_1 / V_2 **or** $1.0 \times 10^5 \times 1.2 \times 10^{-4} / 1.5 \times 10^{-4}$	**C1**
	8.0×10^4 (Pa)	**A1**

Question	Answer	Marks
5(a)	a drawn lens that is thinner at the middle (than at the edges)	**B1**
5(b)(i)	1.8 (cm)	**B1**
5(b)(ii)	vertical arrow of height 2.1 cm drawn 3.0 cm from lens (and labelled O)	**B1**
5(b)(iii)	any **two** from: paraxial ray from tip of object to lens that then refracts through F_2 ray from tip of object, straight through and past optical centre ray from tip of object through F_1 to lens that then emerges paraxially	**B2**
	image drawn from (sensible) intersection to principal axis	**B1**
5(b)(iv)	(linear magnification =) h_I / h_o **or** height of candidate's image / 2.1 (cm)	**C1**
	$1.2 \leqslant$ magnification $\leqslant 1.8$	**A1**
5(b)(v)	real **and** formed by converging light / rays	**B1**

Question	Answer	Marks
6(a)	current / it in AB / the (magnetising) coil produces magnetic field	**B1**
	(relay) switch (between X and Y) closes **or** switch attracted by magnetic field / coil **or** (relay) switch made of iron	**B1**

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Question	Answer	Marks
6(b)	(initially) the current in the relay coil is large **or** relay switch (in relay circuit / relay) is closed **or** heater is connected to a 12 V supply **or** voltage of supply does not change **or** heater is switched on	**B1**
	(as *R* increases) current / magnetic field (in relay coil) decreases	**B1**
	(at $R = R_H$) relay switch (in relay coil) opens **or** magnetic field not strong enough to hold switch closed	**B1**
6(c)(i)	($E =$) Pt **or** 40×2.0 **or** $40 \times 2.0 \times 60$ **or** 80	**C1**
	4800 (J)	**A1**
6(c)(ii)	$I = P / V$ **or** $I = 40 / 12$ **or** $I = 3.3$ (A)	**C1**
	($R =$) V / I **or** $12 / 3.3$	**C1**
	3.6 (Ω)	**A1**

Question	Answer	Marks
7(a)	sinusoidal curved line drawn on Fig. 7.1	**B1**
	at least **one** sinusoidal cycle of period 0.020 (s) on Fig. 7.1	**B1**
	2.5 sinusoidal cycles of period 0.020 (s) **and** amplitude of 300 V on at least one cycle on Fig. 7.1	**B1**
7(b)(i)	it is a temporary magnetic material	**B1**
7(b)(ii)	(alternating / changing) current in primary coil	**B1**
	alternating / changing magnetic field (in core / in secondary coil)	**B1**
	voltage / e.m.f. induced (in secondary coil)	**B1**

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Question	Answer	Marks
7(b)(iii)	(V_S =) $N_S V_P$ / N_P **or** 60 × 300 / 750	**C1**
	24 (V)	**A1**

Question	Answer	Marks
8(a)(i)	2 neutrons **and** 2 protons **and** no other particles present	**B1**
8(a)(ii)	helium **and** nucleus	**B1**
8(b)(i)	138 **cao**	**B1**
8(b)(ii)	88 **cao**	**B1**
8(c)	7.2×10^{16} / 9.0×10^{15} **or** 9.0×10^{15} / 7.2×10^{16} **or** 8(×) **or** 1 / 8(th)	**C1**
	3 half-lives identified **or** 3 × 75 000	**C1**
	2.25×10^5 (years)	**A1**
8(d)(i)	horizontal line in electric field **and** nothing else	**B1**
8(d)(ii)	horizontal line in magnetic field **and** nothing else	**B1**

Question	Answer	Marks
9(a)(i)	(v =) s / t **or** any relevant distance / any relevant time	**C1**
	(v =) $2\pi \times 6.4 \times 10^6$ / (24 × 3600)	**C1**
	470 (m / s)	**A1**

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Question	Answer	Marks
9(a)(ii)	(circumference of) path decreases (with distance from Equator)	**B1**
	rotation period / time for rotation remains constant	**B1**
9(b)(i)	gravitational attraction / field	**B1**
	due to Sun	**B1**
9(b)(ii)	increases **and** decreases **or** fluctuates **or** varies **or** not constant	**B1**
	decreases when moving away from Sun **or** increases when moving towards Sun	**B1**
9(b)(iii)	(at greatest distance) the gravitational potential energy is the greatest	**B1**
	(hence) kinetic energy the smallest (and so speed is the smallest)	**B1**

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